

# Certified Inversion Through the $q \rightarrow 1$ Transition

Signed remainders, optimal truncation,  
and a modular cancellation window

With a shortest-block law for Gaussian multinomials

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## Abstract

The simultaneous limit  $q = e^h \rightarrow 1^+$  and  $x \rightarrow \infty$  reorganizes the transseries of Gaussian binomial coefficients: the finite-product tail enters a dilogarithmic dominant phase, while the Euler product contributes a separate modular exponential. We prove an exact positive-kernel remainder formula for the canonical real interpolation of the central Gaussian binomial. It yields signed first-omitted-term bounds for every  $h, x > 0$ , a fixed-order bound uniform up to  $hx = 0$ , and two-sided rational approximations from a determinate Stieltjes measure.

The main quantitative result identifies the optimally truncated finite-size remainder, including its first relative correction. Its leading value is  $(-1)^{M+1}e^{-2\pi x}/(\pi\sqrt{x})$  when  $M = \pi x + O(1)$  and  $hx$  stays in a compact positive interval. Combining it with the modular term  $e^{-4\pi^2/h}$  gives a stable additive law for inverse errors. Their true balance occurs in a logarithmically shifted window, not exactly at  $hx = 2\pi$ ; even and odd truncations respectively cancel and reinforce. A multinomial extension shows that the shortest block and its multiplicity determine the leading optimal remainder. Complete proofs, reproducible exact-rational and high-precision checks, a repository crosswalk, and a further-research agenda are included.

**Status and scope.** This is a proof-based research manuscript, not an independently refereed or Lean-checked result. The explicit refinements are developed relative to the inspected repository chapter. Classical partial-fraction, modular, moment, and summation machinery is attributed; worldwide priority for the refinements has not been established. No general open conjecture about all transseries is claimed solved.

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# 1 The problem, the contribution, and the boundary of novelty

## 1.1 A concrete gap rather than a generic request for a breakthrough

The PROVEIt companion *Combinatorial Transseries and Their Inverses* already contains a detailed chapter on the singular transition  $q \rightarrow 1$  [1]. Its forward expansion isolates a dilogarithmic phase, a logarithmic amplitude, Bernoulli corrections, and an exact modular factor. It also gives finite-order inverse corrections and an all-order formal inverse formula. Those are starting points here, not new results of this manuscript.

Two distinctions in that chapter are particularly important. Its uniform finite-order error theorem assumes  $\tau = hx \geq \tau_0 > 0$ , and it explicitly leaves uniform matching to  $\tau = 0$  to a different analysis. It also states that the modular factor is not the whole beyond-all-orders structure of the finite-size tail. Consequently, retaining a modular term next to an unspecified algebraic error does not establish which exponential actually controls an inverse approximation. The present paper supplies a positive remainder kernel and uses it to answer that more precise question.

The resulting advances are linked rather than a collection of unrelated coefficient calculations. Positivity makes the divergent expansion enveloping. The same positivity yields a Stieltjes measure and rational enclosures. A large-moment analysis then determines the exact leading optimal remainder, which permits a signed comparison with the modular correction. Finally, inverse-error transport preserves that comparison because the interpolation has a uniformly positive slope.

## 1.2 What is proved here

The principal results are the following. [Theorem 3.2](#) gives an exact signed remainder and a strict first-omitted-term bound. [Theorem 3.3](#) extends fixed-order control to  $0 < hx \leq \tau_1$  in the natural variable  $x$ . [Theorems 5.1](#) and [5.2](#) give a positive Stieltjes representation and convergent two-sided Padé bounds. [Theorem 7.2](#) determines the optimal remainder through relative order  $x^{-1}$ . [Theorems 8.2](#) and [8.3](#) give the two-exponential inverse-error law and its logarithmically shifted cancellation window. [Theorem 9.2](#) extends the leading law to arbitrary fixed positive multinomial block proportions.

These are assertions with proofs below. The numerical checks are separate: they test signs, constants, rational bounds, and nonlinear residuals, but do not replace the arguments. In particular, no unexecuted Lean code is presented as formal verification.

## 1.3 Relation to existing analysis

The partial-fraction expansion of the hyperbolic cotangent and the modular transformation of the Dedekind eta function are classical; convenient precise references are DLMF 4.36.3 and 23.18.5 [3, 4]. Modular representations of general  $q$ -Pochhammer products involving the dilogarithm and Laplace integrals were developed by Zhang [5]. Modern resurgence and summability results for weighted  $q$ -Pochhammer sums appear in Fantini and Rella [6]. The existence of a Borel description of a  $q$ -product is therefore not a novelty claim here.

Likewise, positivity, quadrature, moment determinacy, and Padé convergence belong to classical Stieltjes theory; Simon gives a substantial treatment of its operator and approximation aspects [7]. Here the specific measure, the resulting certificates for this interpolation, and their connection to the inverse cancellation calculation are made explicit.

The sharp parity-sensitive inverse comparison is the main proposed research contribution. A targeted comparison with the cited sources and the pinned repository does not constitute an

exhaustive priority search. The assertions are deliberately formulated so that their correctness can be assessed without accepting a claim that they are unprecedented.

## 2 Exact interpolation and the two small mechanisms

For  $0 < Q < 1$ , write

$$(z; Q)_\infty = \prod_{j=0}^{\infty} (1 - zQ^j).$$

Throughout the central-binomial analysis,

$$h > 0, \quad Q = e^{-h}, \quad q = e^h, \quad x > 0, \quad \tau = hx, \quad t = e^{-\tau}. \quad (2.1)$$

All logarithms of positive quantities are real logarithms. Define

$$P_h = (Q; Q)_\infty, \quad P_h(x) = \frac{(Q; Q)_\infty}{(Q^{x+1}; Q)_\infty}, \quad H_h(x) = e^{hx^2} \frac{P_h(2x)}{P_h(x)^2}, \quad L_h(x) = \log H_h(x). \quad (2.2)$$

The distinction between the constant  $P_h$  and the function  $P_h(x)$  is maintained throughout. At an integer  $n \geq 0$ ,  $P_h(n) = \prod_{j=1}^n (1 - Q^j)$ , and elementary reversal of the factors gives

$$H_h(n) = \left[ \frac{2n}{n} \right]_q.$$

Thus (2.2) specifies an interpolation, rather than assuming that an integer sequence has a unique analytic continuation.

Introduce the exact tail

$$A_h(t) = \sum_{m=1}^{\infty} \frac{t^m}{m(e^{hm} - 1)}. \quad (2.3)$$

Expanding  $-\log(1 - z)$  inside absolutely convergent products proves

$$L_h(x) = hx^2 - \log P_h + A_h(t^2) - 2A_h(t). \quad (2.4)$$

The modular identity supplies the other mechanism:

$$P_h = \sqrt{\frac{2\pi}{h}} \exp\left(-\frac{\pi^2}{6h} + \frac{h}{24}\right) (r_h; r_h)_\infty, \quad r_h = e^{-4\pi^2/h}, \quad (2.5)$$

$$\mathcal{E}(h) = -\log(r_h; r_h)_\infty = \sum_{n=1}^{\infty} \sigma_{-1}(n) r_h^n, \quad \sigma_{-1}(n) = \sum_{d|n} \frac{1}{d}. \quad (2.6)$$

For completeness, apply  $\eta(-1/z) = \sqrt{-iz} \eta(z)$  at  $z = ih/(2\pi)$  and use  $\eta(ih/(2\pi)) = e^{-h/24} P_h$  to obtain (2.5). Expanding each logarithm in the residual Euler product and grouping equal products of indices proves (2.6). In particular,

$$\mathcal{E}(h) = r_h + O(r_h^2), \quad 0 < \mathcal{E}(h) \leq \frac{r_h}{(1 - r_h)^2}. \quad (2.7)$$

The inequality follows from  $\sigma_{-1}(n) \leq n$ .

Define the known phase and combined amplitude

$$S(\tau) = \tau^2 + \frac{\pi^2}{6} + \text{Li}_2(e^{-2\tau}) - 2\text{Li}_2(e^{-\tau}), \quad (2.8)$$

$$G(h, \tau) = \frac{1}{2} \log \left( \frac{h}{2\pi} \coth \frac{\tau}{2} \right). \quad (2.9)$$

Differentiation gives useful elementary identities:

$$S'(\tau) = 2\log(1 + e^\tau), \quad S''(\tau) = \frac{2e^\tau}{1 + e^\tau}, \quad S(0) = 0. \quad (2.10)$$

Hence  $S$  maps  $(0, \infty)$  increasingly onto  $(0, \infty)$ . These phase identities and the formal inverse based on them already occur in the repository [1].

Let the Bernoulli numbers be specified by  $z/(e^z - 1) = \sum_{n \geq 0} B_n z^n / n!$  near zero. For an integer  $M \geq 0$  set

$$B_M(h, x) = \frac{S(\tau)}{h} + G(h, \tau) - \frac{h}{24} + \sum_{k=1}^M \frac{B_{2k} h^{2k-1}}{(2k)!} (\text{Li}_{2-2k}(t^2) - 2 \text{Li}_{2-2k}(t)). \quad (2.11)$$

An empty sum means zero. Our remainder is defined by the exact equation

$$L_h(x) = B_M(h, x) + \mathcal{E}(h) + R_M(h, x). \quad (2.12)$$

The notation  $B_M$  denotes a truncation, not a Bernoulli number when supplied with arguments  $(h, x)$ .

### 3 A global signed remainder formula

**Lemma 3.1** (Finite thermal-kernel remainder). *For every real  $z > 0$  and integer  $M \geq 0$ ,*

$$\frac{1}{e^z - 1} = \frac{1}{z} - \frac{1}{2} + \sum_{j=1}^M \frac{B_{2j} z^{2j-1}}{(2j)!} + \rho_M(z), \quad (3.1)$$

$$\rho_M(z) = (-1)^M 2z^{2M+1} \sum_{k=1}^{\infty} \frac{1}{(2\pi k)^{2M+2} \{1 + (z/(2\pi k))^2\}}. \quad (3.2)$$

*Proof.* The cotangent partial fraction, with its argument halved, yields

$$\frac{1}{e^z - 1} = \frac{1}{z} - \frac{1}{2} + 2z \sum_{k \geq 1} \frac{1}{(2\pi k)^2 + z^2}.$$

In each summand use the finite identity

$$\frac{1}{1+u} = \sum_{j=0}^{M-1} (-u)^j + \frac{(-u)^M}{1+u}.$$

The coefficients agree with the Taylor coefficients at zero; equivalently,

$$\frac{B_{2j}}{(2j)!} = (-1)^{j-1} \frac{2\zeta(2j)}{(2\pi)^{2j}}.$$

The residual sum converges absolutely. This argument uses a *finite* geometric identity at arbitrary positive  $z$ , not an invalid globally convergent Bernoulli expansion.  $\square$

**Theorem 3.2** (Exact signed envelope). *Let*

$$w_m(\tau) = 2e^{-m\tau} - e^{-2m\tau} > 0. \quad (3.3)$$

For all  $h, x > 0$  and all integers  $M \geq 0$ ,

$$R_M(h, x) = (-1)^{M+1} 2h^{2M+1} \sum_{k,m \geq 1} \frac{w_m(\tau) m^{2M}}{(2\pi k)^{2M+2} \{1 + (hm/(2\pi k))^2\}}. \quad (3.4)$$

In particular, with

$$T_{M+1}(h, x) = \frac{|B_{2M+2}| h^{2M+1}}{(2M+2)!} (2 \operatorname{Li}_{-2M}(t) - \operatorname{Li}_{-2M}(t^2)), \quad (3.5)$$

one has the strict inequalities

$$0 < (-1)^{M+1} R_M(h, x) < T_{M+1}(h, x). \quad (3.6)$$

*Proof.* Insert (3.1) into (2.3). The terms  $1/(hm)$  and  $-1/2$  sum to  $\operatorname{Li}_2(t)/h$  and  $\frac{1}{2} \log(1-t)$ . The finite polynomial terms give the polylogarithms in (2.11). The remaining combination is

$$- \sum_{m \geq 1} \frac{w_m(\tau)}{m} \rho_M(hm),$$

which gives (3.4). All absolute sums are finite because  $w_m \leq 2e^{-m\tau}$  and the  $k$ -sum is bounded by a zeta value. Every factor of the unsigned remainder is positive. Replacing  $\{1 + (hm/(2\pi k))^2\}^{-1}$  by 1 strictly increases every summand. Summation and the Bernoulli–zeta identity give (3.5).  $\square$

For even  $M$  the exact logarithm lies below  $B_M + \mathcal{E}$ ; for odd  $M$  it lies above. More explicitly,

$$B_M + \mathcal{E} - T_{M+1} < L_h < B_M + \mathcal{E}, \quad M \text{ even}, \quad (3.7)$$

$$B_M + \mathcal{E} < L_h < B_M + \mathcal{E} + T_{M+1}, \quad M \text{ odd}. \quad (3.8)$$

This is stronger than an unspecified asymptotic  $O$ -term: it is valid before taking any limit and records the sign needed later for cancellation.

### 3.1 Uniform control at the classical endpoint

**Corollary 3.3** (No positive lower cutoff on  $hx$ ). *For fixed  $\tau_1 > 0$  and  $M \geq 0$ , throughout  $0 < hx \leq \tau_1$ ,*

$$|R_M(h, x)| \leq \frac{4\zeta(2M+2)e^{\tau_1}(2M)!}{(2\pi)^{2M+2}x^{2M+1}}. \quad (3.9)$$

Thus the same composite expansion has a uniform fixed-order remainder as  $x \rightarrow \infty$ , even when  $hx \rightarrow 0$ .

*Proof.* For every integer  $p \geq 0$  and  $\tau > 0$ ,

$$\sum_{m \geq 1} m^p e^{-m\tau} \leq e^\tau \int_0^\infty u^p e^{-\tau u} du = e^\tau \frac{p!}{\tau^{p+1}}.$$

Indeed on  $[m, m+1]$ ,  $u^p \geq m^p$  and  $e^{-\tau u} \geq e^{-\tau(m+1)}$ . Apply this inequality with  $p = 2M$  to (3.5), use  $w_m \leq 2e^{-m\tau}$  and  $\tau = hx$ .  $\square$

The endpoint is best understood with  $S$  and  $G$  retained intact. Their small  $\tau$  expansions are

$$S(\tau) = 2(\log 2)\tau + \frac{\tau^2}{2} + \frac{\tau^3}{12} + O(\tau^5), \quad (3.10)$$

$$G(h, \tau) = -\frac{1}{2} \log(\pi x) + O(\tau^2). \quad (3.11)$$

For each fixed  $k \geq 1$ ,

$$\frac{B_{2k} h^{2k-1}}{(2k)!} (\text{Li}_{2-2k}(e^{-2\tau}) - 2 \text{Li}_{2-2k}(e^{-\tau})) \sim \frac{(2^{1-2k} - 2)B_{2k}}{2k(2k-1)x^{2k-1}} \quad (\tau \downarrow 0). \quad (3.12)$$

These are the central-binomial Stirling coefficients. At  $k = 1$  the constant part of the polylogarithm combination cancels the separately retained  $-h/24$ . In fact their sum is

$$-\frac{h}{24} + \frac{h}{12} (\text{Li}_0(t^2) - 2 \text{Li}_0(t)) = -h \frac{e^{2\tau} + 4e^\tau + 1}{24(e^{2\tau} - 1)} = -\frac{1}{8x} - \frac{h\tau^3}{1440} + O(h\tau^5). \quad (3.13)$$

*Remark 3.4* (What endpoint matching does not say). The condition  $hx \rightarrow 0$  alone does not permit dropping all  $q$ -dependent phase corrections. For example  $\tau^2/(2h) = hx^2/2$  can be large even when  $hx$  is small. The uniform statement is (3.9) for the composite expression  $B_M$ , not an assertion that  $H_h(x)$  is relatively close to the classical binomial under only  $hx \rightarrow 0$ . This distinction avoids a common nonuniform interchange of limits.

## 4 Monotonicity and certified inverse transport

**Proposition 4.1** (A slope bound independent of  $h$ ). *The function  $L_h$  is strictly increasing on  $(0, \infty)$ , extends continuously with  $L_h(0) = 0$ , and tends to infinity. For  $x \geq 1$ ,*

$$\mu := 2 \log \frac{3}{2} \leq L'_h(x) \leq S'(hx). \quad (4.1)$$

*Proof.* Differentiate (2.4), justified by exponential convergence on compact positive  $x$ -intervals:

$$L'_h(x) = 2hx + 2h \sum_{m \geq 1} \frac{e^{-hmx} - e^{-2hmx}}{e^{hm} - 1} > 0. \quad (4.2)$$

The product definition gives continuity at zero; the  $hx^2$  term in (2.4) gives divergence at infinity. The inequalities

$$\frac{e^{-hm}}{m} \leq \frac{h}{e^{hm} - 1} \leq \frac{1}{m}$$

follow from  $1 - e^{-z} \leq z \leq e^z - 1$ . Summing the lower estimate in (4.2) and combining the  $2hx$  term with the logarithms gives

$$L'_h(x) \geq 2 \log \frac{e^{h(2x+1)} - 1}{e^{h(x+1)} - 1} \geq 2 \log \frac{2x+1}{x+1}.$$

The second inequality follows because  $(e^u - 1)/u$  increases for  $u > 0$ . For  $x \geq 1$  the last ratio is at least  $3/2$ . The upper estimate sums to  $2hx + 2 \log(1 + e^{-hx}) = S'(hx)$ .  $\square$

Given a positive target  $Y$ , put  $y = \log Y$ . The real inverse exists uniquely when  $Y \geq 1$ . For an inverse  $x_*$  and a trial point  $z$ , both at least one,

$$|z - x_*| \leq \frac{|L_h(z) - y|}{\mu}. \quad (4.3)$$

This follows directly from the mean value theorem. With any rigorous logarithmic enclosure  $\ell_-(z) \leq L_h(z) \leq \ell_+(z)$ , a usable bound is

$$|z - x_*| \leq \frac{\max\{|\ell_-(z) - y|, |\ell_+(z) - y|\}}{\mu}. \quad (4.4)$$

The assumption that the interval of interest lies in  $[1, \infty)$  is explicit; for smaller indices one uses a positive local slope minimum instead.

The signed envelopes (3.7)–(3.8) give immediate choices of  $\ell_{\pm}$ . A bracket  $u < v$  with  $\ell_+(u) < y < \ell_-(v)$  contains the unique inverse. This bracketing method does not require a truncated expression itself to be globally monotone.

For the integer threshold  $\min\{n : H_h(n) \geq Y\}$ , the answer is  $\lceil x_* \rceil$ . A certified bracket determines it when its endpoint ceilings agree. If the bracket crosses an integer, evaluate the exact finite Gaussian product there. Equality must not be decided from a rounded floating-point approximation. These order-theoretic principles already belong to the repository's inversion architecture [1]; the explicit forward envelopes above are the additional input.

## 5 A positive Stieltjes transform and rational enclosures

**Theorem 5.1** (The finite-size correction is Stieltjes). *Fix  $\tau > 0$ . Define, for  $s \geq 0$ ,*

$$\mathcal{D}_{\tau}(s) = 2 \sum_{k,m \geq 1} \frac{w_m(\tau)}{(2\pi k)^2 + sm^2}. \quad (5.1)$$

Then

$$L_h(\tau/h) = \frac{S(\tau)}{h} + G(h, \tau) - \frac{h}{24} + \mathcal{E}(h) - h\mathcal{D}_{\tau}(h^2). \quad (5.2)$$

More precisely,

$$\mathcal{D}_{\tau}(s) = \int_{[0, \infty)} \frac{dv_{\tau}(u)}{1 + su}, \quad (5.3)$$

$$v_{\tau} = 2 \sum_{k,m \geq 1} \frac{w_m(\tau)}{(2\pi k)^2} \delta_{m^2/(2\pi k)^2}, \quad (5.4)$$

$$\mu_j(\tau) := \int u^j dv_{\tau}(u) = \frac{|B_{2j+2}|}{(2j+2)!} (2 \operatorname{Li}_{-2j}(t) - \operatorname{Li}_{-2j}(t^2)). \quad (5.5)$$

The measure is finite and positive, has infinite support, has all moments, and is uniquely determined among measures on  $[0, \infty)$  by those moments.

*Proof.* The case  $M = 0$  of [theorem 3.1](#) gives (5.2). Factoring  $(2\pi k)^2$  from each denominator proves (5.3). Tonelli's theorem evaluates the moments as a product of a zeta value and an exponential power sum, giving (5.5). Positivity is strict on infinitely many support points, for example the points with  $k = 1$ .

Here is a direct determinacy argument. For  $0 < c < 2\pi\tau$ ,

$$\int e^{c\sqrt{u}} dv_{\tau}(u) \leq 4 \sum_{k,m \geq 1} \frac{e^{-m\tau + cm/(2\pi k)}}{(2\pi k)^2} < \infty. \quad (5.6)$$

Also [theorem 3.3](#)'s moment estimate implies  $\mu_j \leq CA^{2j}(2j)!$  for constants depending on  $\tau$ . For any other positive measure with these moments, the monotone expansion of  $\cosh(c\sqrt{u})$  gives



a finite integral for sufficiently small  $c$ . Its symmetric square-root lift to  $\mathbb{R}$  therefore has an exponential moment near zero. The moment generating function of that lift is analytic there and determined by the moments; equality of the analytic transforms, followed by uniqueness of the Fourier transform after exponential tilting, determines the lift and hence the original measure. Equivalently, the Stieltjes Carleman sum  $\sum_j \mu_j^{-1/(2j)}$  diverges.  $\square$

For  $s > 0$ , differentiation under the integral gives

$$(-1)^j \mathcal{D}_\tau^{(j)}(s) = j! \int \frac{u^j \mathrm{d}v_\tau(u)}{(1 + su)^{j+1}} > 0. \quad (5.7)$$

Thus  $\mathcal{D}_\tau$  is completely monotone. All Hankel moment matrices are positive definite: for any nonzero polynomial  $p$ ,  $\int p(u)^2 \mathrm{d}v_\tau(u) > 0$ . The formal expansion is

$$\mathcal{D}_\tau(s) \sim \sum_{j \geq 0} (-1)^j \mu_j(\tau) s^j. \quad (5.8)$$

It is generally divergent at every nonzero  $s$  as a power series. Positivity nevertheless gives convergent rational approximations.

**Theorem 5.2** (Two-sided Padé certificates). *Let  $[L/N]_\tau$  denote the Padé approximant to (5.8), normalized with denominator constant term one. For  $N \geq 1$  and  $s > 0$ ,*

$$[N - 1/N]_\tau(s) < \mathcal{D}_\tau(s) < [N/N]_\tau(s). \quad (5.9)$$

*Both approximants exist, have no pole on  $(0, \infty)$ , and converge pointwise to  $\mathcal{D}_\tau(s)$  as  $N \rightarrow \infty$ . Consequently, replacing  $\mathcal{D}_\tau$  in (5.2) by the upper and lower rational bounds gives respectively lower and upper bounds for  $L_h$ .*

*Proof.* Take the  $N$ -node Gaussian quadrature for  $v_\tau$ . Its nodes are positive, its weights are positive, and it is exact through degree  $2N - 1$ . These facts follow from the positive definite moment form and its degree- $N$  orthogonal polynomial. Applied to  $f_s(u) = (1 + su)^{-1}$ , it is a rational function with numerator degree at most  $N - 1$  and denominator degree  $N$ , whose expansion agrees through  $s^{2N-1}$ . It is therefore  $[N - 1/N]_\tau$ .

The polynomial Hermite interpolant matching  $f_s$  and  $f'_s$  at the Gaussian nodes lies below  $f_s$  on  $[0, \infty)$ : its error has the sign of  $f_s^{(2N)}$  times a squared node polynomial, which is positive. Integrating and using quadrature exactness proves the lower inequality. This argument is valid on each finite interval containing the evaluation point; its global integral is permitted because all moments exist.

For the upper bound use Gauss–Radau quadrature with the prescribed node zero and  $N$  further positive nodes. The latter are the zeros of the degree- $N$  orthogonal polynomial for  $u \mathrm{d}v_\tau(u)$ . The quadrature is exact through degree  $2N$ . The interpolant matches at zero and matches values and derivatives at the other nodes. Its error has the sign of  $f_s^{(2N+1)} u$  times a squared polynomial, which is negative. Thus the interpolant, and after integration the quadrature, is an upper bound. Its rational expression is  $[N/N]_\tau$ . Infinite support makes both inequalities strict.

For convergence, regard either sequence of quadratures as positive discrete measures. Their total mass is  $\mu_0$  and their first moment is eventually  $\mu_1$ , so they are tight. Each higher moment is eventually exact; the next higher exact moment supplies uniform integrability for passing that moment to a weak limit. Every subsequential limit therefore has all moments  $\mu_j$ . Determinacy in theorem 5.1 forces the limit to be  $v_\tau$ . The bounded continuous function  $f_s$  then gives convergence of the quadrature values. Positive nodes ensure the absence of positive poles.  $\square$

**Example 5.3** (Rational certificates at  $t = 1/2$ ). With  $\tau = \log 2$ ,

$$\mu_0 = \frac{5}{36}, \quad \mu_1 = \frac{19}{1215}, \quad [0/1]_\tau(s) = \frac{5/36}{1 + (76/675)s}.$$

Every moment is rational, since negative-integer polylogarithms are rational functions of  $t$ . Hence every finite Padé bound can be built in exact rational arithmetic. The computation in [section 10](#) uses this fact; it does not infer rigorous inequalities from decimal agreement.

## 6 Borel reconstruction and the relevant singularity scale

The measure also exposes the exponential scale without making a claim that the entire modular completion is encoded in the algebraic coefficients. Use the Borel convention

$$\mathcal{B}\left(\sum_{n \geq 0} a_n h^{n+1}\right)(\xi) = \sum_{n \geq 0} \frac{a_n}{n!} \xi^n.$$

For the finite-size series  $F_\tau(h) \sim -h \sum_{j \geq 0} (-1)^j \mu_j h^{2j}$ , its Borel transform is initially

$$\widehat{F}_\tau(\xi) = -2 \sum_{k, m \geq 1} \frac{w_m(\tau)}{(2\pi k)^2} \cos \frac{m\xi}{2\pi k}. \quad (6.1)$$

This converges normally for  $|\operatorname{Im} \xi| < 2\pi\tau$ .

**Proposition 6.1** (Real Borel sum and nearest singularities). *For  $h > 0$ ,*

$$\int_0^\infty e^{-\xi/h} \widehat{F}_\tau(\xi) d\xi = -h \mathcal{D}_\tau(h^2). \quad (6.2)$$

*The transform admits a meromorphic continuation whose possible poles lie in*

$$\{4\pi^2 k \ell \pm 2\pi i k \tau : k \geq 1, \ell \in \mathbb{Z}\} \cup \{4\pi^2 k \ell \pm 4\pi i k \tau : k \geq 1, \ell \in \mathbb{Z}\}. \quad (6.3)$$

*The nearest singularities to the origin are the nonzero simple poles  $\xi = \pm 2\pi i \tau$ .*

*Proof.* On the real  $\xi$ -axis, the absolute value of the double series is bounded by  $2 \sum_{k, m} w_m / (2\pi k)^2 < \infty$ . Fubini and

$$\int_0^\infty e^{-\xi/h} \cos(b\xi) d\xi = \frac{h}{1 + h^2 b^2}$$

prove (6.2). For continuation write

$$C_r(z) = \frac{1}{2} \left( \frac{r e^{iz}}{1 - r e^{iz}} + \frac{r e^{-iz}}{1 - r e^{-iz}} \right).$$

Then

$$\widehat{F}_\tau(\xi) = -2 \sum_{k \geq 1} (2\pi k)^{-2} \{2C_t(\xi/(2\pi k)) - C_{t^2}(\xi/(2\pi k))\}.$$

For sufficiently large  $k$ , the argument lies uniformly in a small neighborhood of zero on any fixed compact set; the remaining  $k^{-2}$  series converges normally. The finite set of small- $k$  summands gives the listed possible poles. At  $\pm 2\pi i \tau$ , only the  $k = 1$ ,  $C_t$  term has a pole, so cancellation is impossible. No listed pole has smaller modulus.  $\square$

Some farther poles in (6.3) can cancel, so that set is not asserted to be an exact divisor. The distance  $2\pi\tau$  from the origin predicts the magnitude  $e^{-2\pi\tau/h} = e^{-2\pi x}$  of a least-term error. It does not imply that a new independent positive-real transseries parameter of that size occurs in the exact function: on the positive axis (6.2) already defines the finite-size contribution unambiguously. The optimal error is the difference between that sum and a finite truncation. The modular term  $\mathcal{E}(h)$  is a separate, exactly specified flat contribution.

This distinction between a truncation error, a Borel singularity, and a separately normalized nonperturbative term is essential. The cancellation studied below is a real approximation-error phenomenon, not by itself a Stokes jump.

## 7 Sharp optimal truncation, with a first correction

In this section  $h \downarrow 0$ ,  $\tau = hx$  belongs to a fixed compact set  $K \subset (0, \infty)$ , and the integer  $M$  satisfies  $|M - \pi x| \leq C$  for a fixed constant  $C$ . Define

$$a = 2\pi x, \quad d = 2M - a, \quad C_1(d) = \frac{d^2}{2} - \frac{d}{2} - \frac{5}{12}, \quad a_x = \frac{e^{-2\pi x}}{\pi\sqrt{x}}. \quad (7.1)$$

Here  $a_x$  is an error scale; it is unrelated to the block lengths introduced later.

**Lemma 7.1** (Uniform large exponential moments). *For integers  $p \geq 1$ , let  $Z_p(\tau) = \sum_{m \geq 1} m^p e^{-m\tau}$ . Then*

$$Z_p(\tau) = p! \sum_{\ell \in \mathbb{Z}} (\tau + 2\pi i \ell)^{-p-1}. \quad (7.2)$$

For  $\tau \in K$ , there is  $0 < \rho_K < 1$  such that, as  $p \rightarrow \infty$ ,

$$Z_p(\tau) = \frac{p!}{\tau^{p+1}} (1 + O_K(\rho_K^p)). \quad (7.3)$$

The same assertion holds for  $Z_{p+j}/Z_p$  with each fixed nonnegative  $j$ , after dividing by the corresponding factorial and power ratio.

*Proof.* The bilateral partial fraction for the derivative of the thermal kernel is

$$\frac{e^{-\tau}}{(1 - e^{-\tau})^2} = \sum_{\ell \in \mathbb{Z}} (\tau + 2\pi i \ell)^{-2}.$$

It is also the  $\text{csch}^2$  partial fraction. Differentiating  $p - 1$  times gives (7.2); all derivative series converge normally on compact positive  $\tau$ -intervals. The  $\ell = 0$  term is the displayed leading term. For the rest, the ratios  $|\tau/(\tau + 2\pi i \ell)|$  are uniformly below one for  $\ell \neq 0$ . Bounding two powers by a summable  $\ell^{-2}$  sequence and the remaining powers by a constant less than one proves (7.3). The same estimate with  $p + j$  proves the final assertion.  $\square$

**Theorem 7.2** (Leading optimal remainder and its relative correction). *Under the compactness and bounded-offset hypotheses above,*

$$R_M(h, x) = (-1)^{M+1} a_x \left( 1 + \frac{C_1(d)}{2\pi x} + O_{K,C}(x^{-2}) \right). \quad (7.4)$$

The first omitted term satisfies

$$T_{M+1}(h, x) = 2a_x \left( 1 + \frac{d^2/2 + d/2 + 1/12}{2\pi x} + O_{K,C}(x^{-2}) \right). \quad (7.5)$$

In particular  $|R_M|/T_{M+1} \rightarrow 1/2$ , uniformly in this regime.

*Proof.* Put  $p = 2M = a + d$ . In (3.4), the contribution with  $k = 1$  and  $w_m$  replaced by  $2e^{-m\tau}$  is

$$V = \frac{4h^{p+1}}{(2\pi)^{p+2}} Z_p(\tau) \mathbb{E}_p f(Y), \quad f(v) = \frac{1}{1+v^2}, \quad Y = \frac{hm}{2\pi}, \quad (7.6)$$

where  $\mathbb{P}_p(m) = m^p e^{-m\tau} / Z_p(\tau)$ .

First control the discarded contributions. Replacing  $e^{-m\tau}$  by  $e^{-2m\tau}$  reduces the leading large moment by  $2^{-p-1}$ , up to a uniform relative error exponentially small in  $p$ . The sum over  $k \geq 2$  is bounded by its unattenuated moment times  $\sum_{k \geq 2} k^{-p-2} = O(2^{-p})$ . The main expectation tends to  $1/2$ , as shown next, so both discarded parts are exponentially small relative to  $V$ . This also justifies keeping a sign when computing the main contribution.

The moment-ratio statement in theorem 7.1 gives

$$\begin{aligned} \mathbb{E}_p(Y - 1) &= \frac{d+1}{a} + O_{K,C}(\rho^p), \\ \mathbb{E}_p(Y - 1)^2 &= \frac{1}{a} + O_{K,C}(a^{-2}), \\ \mathbb{E}_p(Y - 1)^3 &= O_{K,C}(a^{-2}), \quad \mathbb{E}_p(Y - 1)^4 = O_{K,C}(a^{-2}). \end{aligned}$$

Polynomial factors multiplying the exponential errors may be absorbed by increasing  $\rho < 1$ . All derivatives of  $f$  needed here are bounded on  $[0, \infty)$ . Taylor expansion through degree three at one, with the fourth moment bounding the remainder, therefore yields

$$\mathbb{E}_p f(Y) = \frac{1}{2} - racd/2 + 1/4a + O_{K,C}(a^{-2}), \quad (7.7)$$

since  $f'(1) = -1/2$  and  $f''(1) = 1/2$ .

Using (7.3) in the factor before the expectation gives

$$\frac{4h^{p+1}}{(2\pi)^{p+2}} Z_p(\tau) = \frac{2}{\pi} \frac{\Gamma(p+1)}{a^{p+1}} (1 + O_K(\rho^p)).$$

Stirling's formula at  $p = a + d$ , uniformly for bounded  $d$ , reads

$$\frac{\Gamma(a+d+1)}{a^{a+d+1}} = \frac{e^{-a}}{\sqrt{x}} \left( 1 + \frac{d^2/2 + d/2 + 1/12}{a} + O_C(a^{-2}) \right). \quad (7.8)$$

Multiplying (7.7) by this expansion gives precisely  $C_1(d) = d^2/2 - d/2 - 5/12$  and proves (7.4). Without the factor  $f(Y)$ , the same calculation gives (7.5). All omitted exponential errors are smaller than  $a^{-2}$  uniformly on  $K$ .  $\square$

*Remark 7.3* (Choosing the index and choosing a parity). The theorem permits the nearest integer to  $\pi x$ , or the nearest even or odd integer. An offset bounded independently of  $x$  changes only the relative correction, not the leading exponential or its prefactor. The quadratic polynomial in (7.4) also distinguishes minimizing the actual remainder from minimizing just the first omitted term. At the continuous-offset level their first correction has its minimum at  $d = 1/2$  and  $d = -1/2$ , respectively. This observation does not assert a global discrete minimum for every finite  $h, x$ .

*Remark 7.4* (A regime not covered by this theorem). Fixed  $h$  with  $x \rightarrow \infty$  forces  $\tau \rightarrow \infty$  and is not covered by the compactness hypothesis. The power-sum saddle then need not be approximated uniformly by its continuous gamma analogue. The fixed-order signed bounds remain valid, but the sharp prefactor in (7.4) is not asserted in that different limiting regime.

## 8 The two-exponential law for inverse error

### 8.1 Local inverse approximants

Fix an exact inverse point by setting  $y = L_h(x)$ . The truncation order is held fixed while the approximate equation is solved. Let  $\widehat{x}_M$  solve

$$B_M(h, \widehat{x}_M) = y \quad (8.1)$$

near  $x$ ; this approximation omits the modular completion. Let  $\widetilde{x}_M$  solve the corrected equation

$$B_M(h, \widetilde{x}_M) + \mathcal{E}(h) = y. \quad (8.2)$$

These are local roots; no global invertibility of every truncation is needed. The next theorem proves their existence and uniqueness in a neighborhood containing the relevant exponentially small displacement.

**Lemma 8.1** (Derivative control near optimal truncation). *Under the hypotheses of [theorem 7.2](#), uniformly on a sufficiently small fixed neighborhood of  $x$ ,*

$$L'_h(z) = S'(hz) + O_K(h), \quad |\partial_z^j R_M(h, z)| = O_{K,C}(a_x) \quad (j = 0, 1, 2), \quad (8.3)$$

and  $\partial_z B_M(h, z) = S'(hz) + O_{K,C}(h) > 0$  for sufficiently small  $h$ .

*Proof.* Differentiate (5.2) along  $\tau = hz$ . The derivative of its amplitude is  $O_K(h)$ ; the derivative of  $-h\mathcal{D}_\tau(h^2)$  is  $O_K(h^2)$ , by exponential domination of each differentiated weight. This proves the first assertion. In (3.4),  $z$  enters only through  $w_m(hz)$ . Each derivative adds at most a fixed power of  $hm$ , and the derivatives of  $w_m$  are bounded by fixed multiples of  $h^j m^j e^{-hmz}$  on the stated neighborhood. The proof of [theorem 7.2](#), with  $p$  replaced by  $p + j$ , bounds the resulting moments by a constant times  $a_x$ . Their denominator factors are positive and at most one. Finally differentiate (2.12). The modular term is independent of  $z$ , and  $S'$  has a positive compact minimum.  $\square$

**Theorem 8.2** (Stable additive inverse-error law). *Let  $\tau = hx \in K \Subset (0, \infty)$  and  $|M - \pi x| \leq C$  as  $h \downarrow 0$ . The local roots (8.1) and (8.2) exist and are unique in a fixed small neighborhood of  $x$ . With  $\sigma_M = (-1)^{M+1}$ ,*

$$\boxed{\widehat{x}_M - x = \frac{r_h + \sigma_M a_x}{S'(\tau)} + O_{K,C}(h(r_h + a_x))}, \quad (8.4)$$

$$\widetilde{x}_M - x = \frac{\sigma_M a_x}{S'(\tau)}(1 + O_{K,C}(h)). \quad (8.5)$$

*The first formula is an additive estimate, valid even when its two leading contributions nearly cancel.*

*Proof.* At  $x$ , the residual for (8.1) is  $B_M(h, x) - y = -\mathcal{E}(h) - R_M(h, x)$ . By [theorem 8.1](#) the derivative is bounded below on a small fixed interval. Evaluating at points at distance  $C_0(r_h + a_x)$  on each side, with  $C_0$  sufficiently large, brackets a zero; strict positivity of the derivative proves uniqueness. The mean value theorem then gives

$$\widehat{x}_M - x = \frac{\mathcal{E}(h) + R_M(h, x)}{\partial_z B_M(h, \xi)}$$

for an intermediate  $\xi$ . Since  $\xi - x = O(r_h + a_x)$ , the denominator is  $S'(\tau) + O(h)$ , uniformly. Insert (2.7) and (7.4). Here  $x^{-1} = O_K(h)$  and  $r_h^2$  is absorbed by  $hr_h$ . This proves (8.4), with an error proportional to the sum of the magnitudes rather than to their possibly vanishing difference. For the corrected equation the residual is only  $-R_M$  and the same proof applies.  $\square$

If  $\tau$  remains in a compact subinterval of  $(0, 2\pi)$ , then  $a_x/r_h \rightarrow \infty$ , so optimal finite-size truncation error dominates. If it remains in a compact subinterval of  $(2\pi, \infty)$ , the modular term dominates. At the nominal crossing  $\tau = 2\pi$ , however,

$$\frac{a_x}{r_h} = \frac{1}{\pi\sqrt{x}} = \frac{\sqrt{h}}{\pi\sqrt{2\pi}} \rightarrow 0. \quad (8.6)$$

Equal exponential actions do not mean equal errors; their algebraic prefactors matter.

## 8.2 The logarithmically shifted window

Define the exact balance coordinate

$$s = s(h, x) := 2\pi x - \frac{4\pi^2}{h} + \frac{1}{2} \log x + \log \pi. \quad (8.7)$$

It is chosen so that

$$\frac{a_x}{r_h} = e^{-s} \quad (8.8)$$

holds identically, not merely asymptotically.

**Theorem 8.3** (Parity-sensitive cancellation window). *Suppose  $s$  stays in a fixed bounded interval and  $|M - \pi x| \leq C$ . Then*

$$\boxed{\frac{S'(hx)}{r_h}(\widehat{x}_M - x) = 1 + (-1)^{M+1}e^{-s} + O_C(h).} \quad (8.9)$$

For even  $M$ , the leading profile is  $1 - e^{-s}$ ; for odd  $M$ , it is  $1 + e^{-s}$ . Let  $x_c(h)$  be the unique positive solution of  $s(h, x) = 0$ . Then

$$x_c(h) = \frac{2\pi}{h} + \frac{\log h}{4\pi} - \frac{\log(\pi\sqrt{2\pi})}{2\pi} + O(h|\log h|), \quad (8.10)$$

$$hx_c(h) = 2\pi + \frac{h \log h}{4\pi} - \frac{h \log(\pi\sqrt{2\pi})}{2\pi} + O(h^2|\log h|). \quad (8.11)$$

At  $x = x_c(h)$  and even  $M$ ,

$$\frac{S'(hx)}{r_h}(\widehat{x}_M - x) = -\frac{C_1(2M - 2\pi x)}{2\pi x} + O_C(h^2). \quad (8.12)$$

In particular the uncorrected inverse error is  $O(hr_h)$  there.

*Proof.* Bounded  $s$  implies  $hx \rightarrow 2\pi$ , as follows immediately from (8.7); in particular the compactness assumptions of [theorem 8.2](#) hold. Divide (8.4) by  $r_h/S'(hx)$  and use (8.8). The function  $s(h, x)$  strictly increases from  $-\infty$  to  $+\infty$  in  $x$ , giving the unique center.

For its expansion write  $x = 2\pi/h + u$ . The defining equation becomes

$$2\pi u + \frac{1}{2} \log(2\pi/h) + \log \pi + \frac{1}{2} \log \left(1 + \frac{hu}{2\pi}\right) = 0.$$

It first gives  $u = O(|\log h|)$  and then (8.10); multiplication by  $h$  proves (8.11).

At  $s = 0$ ,  $a_x = r_h$ . For even  $M$ , [theorem 7.2](#) and (2.7) give

$$\mathcal{E}(h) + R_M(h, x) = -r_h \frac{C_1(d)}{2\pi x} + O_C(h^2 r_h).$$

The root displacement is already  $O(hr_h)$ . Dividing its residual by  $S'(hx) + O(h)$  changes the normalized result only by  $O(h^2)$ , which proves (8.12).  $\square$

The gain  $O(r_h) \rightarrow O(hr_h)$  is an algebraic gain at the same exponential action. It is not a proof that an entire exponential sector disappears. The dependence on  $2M - 2\pi x$  also explains why the refined error oscillates with integer truncation choices instead of varying monotonically in  $h$ .

### 8.3 An exact cancellation point, not only a formal one

**Proposition 8.4** (Unique exact cancellation for fixed even order). *For every fixed  $h > 0$  and every even integer  $M \geq 2$ , there is a unique  $x_{h,M} > 0$  such that*

$$B_M(h, x_{h,M}) = L_h(x_{h,M}). \quad (8.13)$$

Choose even  $M = M(h)$  with  $|M - \pi x_c(h)| \leq C$ . Then, with  $d_c = 2M - 2\pi x_c(h)$ ,

$$x_{h,M} = x_c(h) + \frac{C_1(d_c)}{4\pi^2 x_c(h)} + O_C(h^2). \quad (8.14)$$

*Proof.* Let  $U_M = (-1)^{M+1} R_M > 0$ . For fixed  $h, M$ , every  $w_m(hx)$  strictly decreases in  $x$ , since  $\partial_x w_m = -2hm(e^{-hmx} - e^{-2hmx}) < 0$ . Therefore (3.4) makes  $U_M$  continuous and strictly decreasing. It tends to zero at infinity. For  $M \geq 1$ , its  $k = 1$  part tends to infinity at zero: for large  $m$  the polynomial factor  $m^{2M}/(1 + (hm/(2\pi))^2)$  is bounded below by a positive multiple of  $m^{2M-2}$ , and  $w_m(hx) \rightarrow 1$  for each  $m$ . Monotone convergence then gives divergence. Thus  $U_M = \mathcal{E}(h)$  has exactly one solution. For even  $M$  this is (8.13).

Under the bounded-offset choice, the window theorem and monotonicity first put the solution at  $s = O(h)$ . Use (7.4) there to get

$$s(h, x_{h,M}) = \log \left( 1 + \frac{C_1(2M - 2\pi x_{h,M})}{2\pi x_{h,M}} + O_C(h^2) \right) = \frac{C_1(d_c)}{2\pi x_c} + O_C(h^2).$$

Here moving  $x$  by  $O(h)$  changes the displayed coefficient by  $O(h^2)$ . Since  $\partial_x s = 2\pi + 1/(2x)$ , expansion around  $x_c$  proves (8.14).  $\square$

There is no corresponding zero for an odd truncation: then  $\mathcal{E} + R_M > 0$  everywhere. The exact cancellation point concerns the logarithmic forward residual and consequently any local inverse built at that target. It does not require the truncated expression to be globally invertible.

## 9 Gaussian multinomials and the shortest-block law

Let  $b \geq 2$  be the number of blocks, let  $\alpha = (\alpha_1, \dots, \alpha_b)$  be fixed positive real numbers, and put

$$A = \sum_{i=1}^b \alpha_i, \quad c = \frac{1}{2} \left( A^2 - \sum_{i=1}^b \alpha_i^2 \right), \quad \alpha = \min_i \alpha_i, \quad m_* = \#\{i : \alpha_i = \alpha\}. \quad (9.1)$$

Define

$$H_{h,\alpha}(x) = e^{hc x^2} \frac{P_h(Ax)}{\prod_i P_h(\alpha_i x)}, \quad L_{h,\alpha}(x) = \log H_{h,\alpha}(x). \quad (9.2)$$

When the block sizes  $\alpha_i x$  are integers, this is the usual Gaussian multinomial at  $q = e^h$ . Real positive block lengths specify its interpolation. The central case is  $b = 2$  and  $(\alpha_1, \alpha_2) = (1, 1)$ .

The exact phase and amplitude are

$$S_\alpha(\tau) = c\tau^2 + (b-1)\frac{\pi^2}{6} + \text{Li}_2(e^{-A\tau}) - \sum_i \text{Li}_2(e^{-\alpha_i \tau}), \quad (9.3)$$

$$G_\alpha(h, \tau) = \frac{1}{2} \log \left\{ \left( \frac{h}{2\pi} \right)^{b-1} \frac{1 - e^{-A\tau}}{\prod_i (1 - e^{-\alpha_i \tau})} \right\}. \quad (9.4)$$



Replace the corresponding phase, amplitude, and polylogarithm combination in (2.11) to define

$$B_M^\alpha(h, x) = \frac{S_\alpha(\tau)}{h} + G_\alpha(h, \tau) - \frac{(b-1)h}{24} + \sum_{k=1}^M \frac{B_{2k} h^{2k-1}}{(2k)!} \left\{ \text{Li}_{2-2k}(e^{-A\tau}) - \sum_i \text{Li}_{2-2k}(e^{-\alpha_i \tau}) \right\}. \quad (9.5)$$

Define  $R_M^\alpha$  by

$$L_{h,\alpha}(x) = B_M^\alpha(h, x) + (b-1)\mathcal{E}(h) + R_M^\alpha(h, x). \quad (9.6)$$

**Proposition 9.1** (Positive kernel and uniform inverse slope). *All signed-envelope and Stieltjes conclusions above hold for the multinomial on replacing  $w_m$  by*

$$w_m^\alpha(\tau) = \sum_i e^{-\alpha_i \tau m} - e^{-A\tau m} > 0. \quad (9.7)$$

The phase slope is

$$S'_\alpha(\tau) = \sum_i \alpha_i \log \frac{e^{A\tau} - 1}{e^{\alpha_i \tau} - 1} > 0. \quad (9.8)$$

For  $x \geq 1/\alpha$ ,

$$0 < \mu_\alpha := \sum_i \alpha_i \log \frac{A + \alpha}{\alpha_i + \alpha} \leq L'_{h,\alpha}(x) \leq S'_\alpha(hx). \quad (9.9)$$

*Proof.* The exact logarithm is  $hcx^2 + (1-b)\log P_h + A_h(e^{-A\tau}) - \sum_i A_h(e^{-\alpha_i \tau})$ . Inserting the thermal remainder gives exactly (3.4) with the new weight. It is positive, and is bounded above by  $be^{-\alpha\tau m}$ ; every convergence, moment, and quadrature argument therefore applies. The modular multiplicity is  $b-1$ .

Differentiating the phase and using  $A = \sum_i \alpha_i$  and  $2c = A^2 - \sum_i \alpha_i^2$  proves (9.8). For the exact logarithm the positive derivative correction is

$$h \sum_{m \geq 1} \sum_i \alpha_i \frac{e^{-\alpha_i h x m} - e^{-A h x m}}{e^{hm} - 1}.$$

The same lower thermal bound as in theorem 4.1, combined with the quadratic derivative, yields

$$L'_{h,\alpha}(x) \geq \sum_i \alpha_i \log \frac{e^{h(Ax+1)} - 1}{e^{h(\alpha_i x+1)} - 1} \geq \sum_i \alpha_i \log \frac{Ax+1}{\alpha_i x+1}.$$

Every last ratio increases with  $x$ . Evaluating at  $x = 1/\alpha$  gives the lower bound. The upper thermal bound gives the phase slope.  $\square$

**Theorem 9.2** (Shortest-block universality). *Fix  $\alpha$  and  $K \in (0, \infty)$ . Suppose  $h \downarrow 0$ ,  $hx \in K$ , and  $|M - \pi\alpha x| \leq C$ . Then*

$$\boxed{R_M^\alpha(h, x) = (-1)^{M+1} \frac{m_*}{2\pi\sqrt{\alpha x}} e^{-2\pi\alpha x} (1 + O_{K,C,\alpha}(h)).} \quad (9.10)$$

For a target  $y = L_{h,\alpha}(x)$ , the local root of  $B_M^\alpha(h, \hat{x}) = y$  satisfies

$$\begin{aligned} \hat{x} - x &= \frac{(b-1)r_h + (-1)^{M+1}a_{\alpha,x}}{S'_\alpha(hx)} \\ &\quad + O_{K,C,\alpha}(h\{r_h + a_{\alpha,x}\}), \quad a_{\alpha,x} := \frac{m_* e^{-2\pi\alpha x}}{2\pi\sqrt{\alpha x}}. \end{aligned} \quad (9.11)$$



The normalized cancellation coordinate is

$$s_\alpha = 2\pi\alpha x - \frac{4\pi^2}{h} + \frac{1}{2} \log(\alpha x) + \log \frac{2\pi(b-1)}{m_*}. \quad (9.12)$$

On bounded  $s_\alpha$ -intervals the profile is again  $1 + (-1)^{M+1}e^{-s_\alpha}$ , after normalizing the inverse error by  $S'_\alpha(hx)/((b-1)r_h)$ .

*Proof.* In the power moment of (9.7), the  $m_*$  shortest blocks contribute  $m_*Z_p(\alpha\tau)$ . Every other block contributes a relative factor  $(\alpha/\alpha_i)^{p+1}$ , up to an exponentially small relative error; the subtracted total block contributes  $(\alpha/A)^{p+1}$ . The fixed positive gaps make these negligible. The  $k \geq 2$  remainder is again exponentially small in  $p$ .

Apply the calculation (7.6)–(7.7) with  $a = 2\pi\alpha x$  and the leading weight multiplicity  $m_*$  instead of two. The prefactor before its limiting expectation  $1/2$  is  $(m_*/\pi)\Gamma(p+1)/a^{p+1}$ , giving (9.10). The same differentiated-moment estimates and [theorem 9.1](#) justify local inverse transport exactly as in [theorem 8.2](#). Finally  $a_{\alpha,x}/((b-1)r_h) = e^{-s_\alpha}$  is an identity, proving the profile.  $\square$

This law says more than that a multinomial has a similar kind of expansion. At optimal order, the longest and intermediate blocks disappear from the leading finite-size error; the shortest block controls the exponential action, and its multiplicity controls the prefactor. The remaining blocks still influence the phase slope and the modular multiplicity. Uniformity as two shortest lengths coalesce is a different, explicitly open problem below.

## 10 Reproducible checks and an implementation route

### 10.1 What the supplied program actually verifies

The package includes the script `verify.py` and its recorded output, `results.json`, in the `verification` directory. The script uses exact rational arithmetic for Padé coefficients and Hankel determinants, and variable high-precision arithmetic for the analytic checks. It checks positive remainders against the direct thermal series, compares optimal remainder constants, checks the parity-even inverse window with a nonlinear residual, and checks three multinomial examples. All included assertions passed in the recorded run.

The analytic computations use `mpmath`, not outward-rounded interval arithmetic. Thus the output is evidence of reproducibility and error localization, not a certified numerical proof of every reported decimal. The inequalities are justified by the preceding theorems. A production certifier must combine them with interval evaluations of elementary functions, polylogarithms, and the modular product.

### 10.2 Signed remainder checks

The direct reference value sums

$$-\sum_{m \geq 1} \frac{w_m(\tau)}{m} \left\{ \frac{1}{e^{hm} - 1} - \frac{1}{hm} + \frac{1}{2} \right\}, \quad (10.1)$$

not the Bernoulli expansion being tested. Subtraction from the truncated expression checks both the residual sign and its magnitude. A truncated positive double sum is separately compared using the analytic tail bounds in [section A](#).

| $h$   | $x$ | $M$ | $ R_M $                       | $ R_M /T_{M+1}$ |
|-------|-----|-----|-------------------------------|-----------------|
| 0.4   | 2.5 | 0   | $3.3262523812 \cdot 10^{-2}$  | 0.9905105051    |
| 0.4   | 2.5 | 1   | $3.1866855465 \cdot 10^{-4}$  | 0.9567940861    |
| 0.1   | 10  | 3   | $1.1693752480 \cdot 10^{-10}$ | 0.9861299427    |
| 0.001 | 10  | 4   | $1.6447426745 \cdot 10^{-12}$ | 0.9779321605    |
| 1     | 3   | 6   | $5.4858162652 \cdot 10^{-9}$  | 0.6822534171    |

Table 1: Direct thermal-series remainders obey the signed first-omitted-term bound. The fourth row probes  $hx = 0.01$ , outside any fixed lower cutoff such as  $hx \geq 0.1$ .

| $\tau$ | $h$   | $M$ | $ R_M /a_x$  | $ R_M /T_{M+1}$ | $1 + C_1(d)/(2\pi x)$ |
|--------|-------|-----|--------------|-----------------|-----------------------|
| 1      | 0.5   | 6   | 0.9999174546 | 0.5015392405    | 1.0021411494          |
| 1      | 0.25  | 13  | 0.9822109593 | 0.4742323085    | 0.9811311007          |
| 1      | 0.125 | 25  | 0.9950194354 | 0.4976484589    | 0.9950525697          |
| 6      | 0.5   | 38  | 0.9930042272 | 0.4928088283    | 0.9928846141          |
| 6      | 0.25  | 75  | 1.0019534997 | 0.5009693722    | 1.0019809567          |
| 6      | 0.125 | 151 | 0.9982247493 | 0.4985013113    | 0.9982182871          |

Table 2: Optimal orders use the nearest integer to  $\pi x$ . The bounded rounding offset explains nonmonotonicity between rows. The last column is a prediction, not a fitted constant.

### 10.3 The sharp prefactor and half-next-term rule

The relatively small value  $x = 2$  in the first row is not in a very deep asymptotic regime; the refined prediction need not improve the leading one at every such point. At larger  $x$  the explicit correction consistently tracks the bounded-offset behavior. No monotonic rate of convergence in  $h$  is claimed.

### 10.4 The inverse cancellation window

For each row of [table 3](#), the script solves the scalar balance coordinate for  $x$ , chooses the nearest even  $M$  to  $\pi x$ , forms the exact thermal-series target  $y = L_h(x)$ , and takes one Newton displacement for the truncated inverse equation. It then reevaluates the *nonlinear* equation at the displaced point. The normalized residual magnitudes in the recorded run are below  $1.1 \cdot 10^{-44}$ ,  $1.3 \cdot 10^{-70}$ , and  $1.1 \cdot 10^{-51}$  for  $h = 0.5, 0.25, 0.125$ , respectively. These are floating-point residuals, not formal root certificates.

| $h$   | $s$ | $M$ | $S'(hx) \delta/r_h$ | $1 - e^{-s}$  |
|-------|-----|-----|---------------------|---------------|
| 0.5   | 0   | 38  | -0.0002188003       | 0             |
| 0.25  | -1  | 78  | -1.7244251297       | -1.7182818285 |
| 0.25  | 0   | 78  | 0.0031005179        | 0             |
| 0.25  | 1   | 78  | 0.6329037362        | 0.6321205588  |
| 0.125 | 0   | 156 | -0.0006764509       | 0             |

Table 3: Normalized Newton inverse displacements in the parity-even window. The first relative correction predicts the nonzero small values at  $s = 0$ .

The computed centers are

$$x_c(0.5) = 12.1852197127 \dots, \quad x_c(0.25) = 24.6953774093 \dots, \quad x_c(0.125) = 49.7723472861 \dots$$

For  $h = 0.25$ , the refined normalized prediction at  $s = 0$  is 0.0031306279, compared with the observed 0.0031005179. This tests the polynomial  $C_1(d)$  independently of the leading zero.

## 10.5 Rational enclosures and multinomial checks

At  $t = 1/2$  and  $s = 1/4$ , a direct evaluation gives  $\mathcal{D}_{\log 2}(1/4) = 0.1354393270783473107 \dots$

| $N$ | $[N - 1/N](1/4)$  | $[N/N](1/4)$      | Width                   |
|-----|-------------------|-------------------|-------------------------|
| 1   | 0.135086455331412 | 0.135507348193646 | $4.20893 \cdot 10^{-4}$ |
| 2   | 0.135420747012517 | 0.135445883566755 | $2.51366 \cdot 10^{-5}$ |
| 3   | 0.135436670206401 | 0.135440575785848 | $3.90558 \cdot 10^{-6}$ |
| 4   | 0.135438702706998 | 0.135439673658955 | $9.70952 \cdot 10^{-7}$ |
| 5   | 0.135439129751585 | 0.135439449154842 | $3.19403 \cdot 10^{-7}$ |

Table 4: Decimal displays of exact rational Padé bounds. The JSON output contains the full rational numbers; rounded table entries themselves are not outward-rounded enclosures.

For  $h = 0.2$  and  $x = 10$ , the ratio of the positive multinomial remainder to the shortest-block prediction is 1.0053351399 for both  $(1, 2, 3)$  and  $(1, 1, 2)$ , with orders  $M = 31$ , and 0.9958300329 for  $(2, 2, 3)$ , with  $M = 63$ . The first two ratios agree to the displayed precision although their shortest-block multiplicities differ; the predicted multiplicity was divided out before comparison.

## 10.6 How to choose an order without knowing the inverse

In the compact crossover regime, first solve the zeroth-order composite equation  $B_0(h, z_0) = y$  locally. The exact residual is  $\mathcal{E}(h) + R_0 = O_K(h)$ , while its derivative is bounded below, so  $z_0 - x = O_K(h)$ . Choosing  $M$  to be the nearest integer of the desired parity to  $\pi z_0$  therefore ensures  $|M - \pi x| = O(1)$ , as required by the sharp theorem. This is stronger than using only the phase center, whose missing logarithmic amplitude would cause an  $O(|\log h|)$  index displacement.

For ordinary accuracy, solve with the modular completion retained and use signed or Padé enclosures. Deliberately omitting it to exploit cancellation is a specialized diagnostic, not the default recommendation for a robust numeric inverse. A certified implementation should adapt the order using the actual remainder enclosure rather than rely on asymptotic balance alone.

## 11 Repository integration and proof-assistant boundaries

The inspected snapshot is the full commit 20301601562cf84173f2b503b0dc9c3fbf21cd0c. The source chapter is in

```
Analysis/FabiusFunction/docs/
semi-formalized-research-frontiers/drafts/
series-and-transseries/
Combinatorial_Transseries_Inverses/
Combinatorial_Transseries_Inverses.tex.
```

The following crosswalk prevents a formal coefficient identity from being silently upgraded into an analytic theorem.

| Repository item                                                   | Already present there                                                                         | Added here                                                                                            |
|-------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| <code>t3:prop:double-A</code>                                     | Fixed-order tail expansion for $\tau \geq \tau_0 > 0$ .                                       | Exact sign, explicit next-term majorant, and uniform endpoint estimate (3.9).                         |
| <code>t3:eq:modularP</code> and <code>t3:eq:modularsectors</code> | Exact modular identity and divisor-sum completion.                                            | Quantitative competition with the optimal finite-size error; no claim to a new modular identity.      |
| <code>t3:eq:Sphase</code> and <code>t3:eq:Sderivative</code>      | Dilogarithmic phase and elementary positive derivative.                                       | Uniform lower bound for the <i>exact</i> inverse slope and exponentially accurate residual transport. |
| <code>t3:eq:crossover-all</code>                                  | All formal finite-order inverse coefficients and fixed-order asymptotics.                     | Sharp optimal-order remainder, parity profile, and exact cancellation location.                       |
| Canonical inverse certification framework                         | Residual brackets and order-theoretic threshold principles, with stated formalization scopes. | Explicit signed and rational analytic forward envelopes suitable as inputs to that framework.         |

A useful formalization can be modular. The finite identity in [theorem 3.1](#) after the coth expansion is supplied is algebraic. Positivity and finite-cutoff monotonicity of the remainder kernel are straightforward ordered-field statements. Summability and passage to the infinite sums form the analytic layer. The residual transport theorem then uses only a derivative lower bound and a root-existence bracket.

The hardest new formal components are the bilateral large-moment identity, its uniform exponential estimate, the moment expansion with a bounded integer offset, and the quadrature construction. They should not be bundled into one opaque “transseries theorem”. The repository’s existing distinction between formal identities, conditional analytic lemmas, and assembled asymptotic statements is the appropriate design discipline [2].

No repository files were modified for this deliverable. The present proofs were not compiled in Lean, and no external repository theorem was accepted solely because a README called it proved. The argument uses the printed classical identities and reproduces the mathematical derivations needed for its own claims.

## 12 Further research questions and concrete next targets

The following are proposed problems, not results claimed in the present paper. Each is narrow enough to expose a verifiable output.

- Q1. Sharp optimal remainders uniformly down to  $hx = 0$ .** The fixed-order bound is already uniform there. Can the prefactor and  $C_1(d)$  in [theorem 7.2](#) be made uniform for  $0 < hx \leq \tau_1$ , assuming only  $x \rightarrow \infty$ ? The exact power-sum identity suggests that a strengthened version should be possible, but all differentiated and discarded-weight estimates must be uniform simultaneously.
- Q2. The discrete-saddle transition at large  $hx$ .** The relevant moment distribution has center approximately  $2\pi/h$  and width approximately  $\sqrt{2\pi x}/(hx)$ . Analyze the transition where this width is bounded, rather than large. A theta-modulated prefactor, depending on the fractional position of the saddle, is a concrete candidate. This would connect the present double-scaling theorem to fixed- $q$  asymptotics without an unjustified continuum replacement.
- Q3. Higher cancellation corrections and parity optimization.** Compute the  $x^{-2}$  and  $x^{-3}$  relative corrections to (7.4), with uniform remainders, and propagate them to  $x_{h,M}$ . The

dependence on integer rounding must remain explicit. A practical goal is a certified rule selecting the best of two nearby even orders from the first few correction polynomials.

- Q4. Coalescing shortest blocks.** For multinomials with  $\alpha_2 - \alpha_1 = O(1/x)$ , a second block is no longer exponentially negligible in the large moment. Derive a uniform replacement for the discrete multiplicity  $m_*$ , with a continuous transition profile in  $x(\alpha_2 - \alpha_1)$ . Fixed block proportions in [theorem 9.2](#) do not cover this problem.
- Q5. Which signed  $q$ -product quotients have a positive kernel?** For a general finite quotient, the weight is an exponential polynomial  $\sum_j c_j e^{-a_j u}$ . Characterize useful, algorithmically testable conditions ensuring nonnegativity for every  $u > 0$ . Majorization and Laplace transform ordering are possible approaches, but neither is assumed to give a complete criterion.
- Q6. Quantitative Padé convergence in joint limits.** [Theorem 5.2](#) proves convergence for fixed  $\tau, s$ . Obtain computable rates that remain useful as  $\tau \rightarrow 0, s \rightarrow 0$ , and  $N \rightarrow \infty$  together. The measure has an unbounded support and factorial moments, so compact-support Padé estimates cannot be imported unchanged.
- Q7. An interval-arithmetic inverse and exact integer thresholds.** Implement outward-rounded evaluation of the signed kernel or Padé bounds, including modular-product tails. Prove termination of a bracket-refinement algorithm away from exact thresholds, and use exact finite products to resolve the equality cases. The supplied script is a test harness, not this certifier.
- Q8. Complex  $h$ , actual Stokes data, and inverse resurgence.** Determine the uncanceled divisor of [\(6.3\)](#), compute lateral continuations where necessary, and propagate their differences through compositionally inverted functions. Such a theory should explicitly separate Stokes jumps from the real truncation-error cancellation established here.
- Q9. Normalization and flat-completion uniqueness.** The algebraic series cannot by itself determine the modular term. Identify a minimal combination of a  $q$ -difference equation, growth conditions, and a boundary normalization that uniquely recovers the chosen Gaussian interpolation among functions sharing the same algebraic expansion.
- Q10. Shifted quotients such as  $q$ -Catalan families.** Block lengths of the form  $\alpha_i x + \beta_i$  introduce fixed shifts in the exponential weights. Determine the leading optimal remainder and cancellation window including the shift-dependent prefactor. This is not obtained by simply replacing  $x$  with  $x + 1$  in the central-binomial answer.
- Q11. Theta-weighted finite-field enumeration.** Extend the positive-kernel viewpoint to the theta-modulated Galois and flag families already studied in the companion. Establish precisely when a positive measure survives the additional lattice sum, and when oscillatory or signed kernels require a different certification method.
- Q12. A composable Lean certificate library.** Formalize a finite geometric remainder interface, positive summation bounds, residual-to-root transport, and a symbolic truncation checker before attempting full Borel or quadrature formalization. A concrete first milestone is a machine-checked finite-cutoff version of [\(3.4\)](#) with an explicit analytic tail supplied as a hypothesis.

## 13 Conclusion

The singular  $q \rightarrow 1$  transition is not adequately described by a formal inverse expansion plus an isolated modular term. Its finite-size Bernoulli series has a structured, signed remainder of its

own. A positive thermal kernel makes that structure explicit and gives three kinds of control: fixed-order bounds uniform toward the classical endpoint, rational Stieltjes enclosures, and sharp optimal-order asymptotics.

The central quantitative conclusion is the stable inverse-error law

$$\widehat{x}_M - x = \frac{e^{-4\pi^2/h} + (-1)^{M+1}e^{-2\pi x}/(\pi\sqrt{x})}{2\log(1 + e^{hx})} + O\left(h\left\{e^{-4\pi^2/h} + \frac{e^{-2\pi x}}{\pi\sqrt{x}}\right\}\right),$$

under the precisely stated compact double-scaling hypotheses. Its balance coordinate includes a logarithmic prefactor shift, its sign depends on truncation parity, and its refined cancellation location is explicit. The multinomial version shows that the shortest block and its multiplicity control the leading finite-size error, while the full composition controls the inverse slope.

These are delimited analytic refinements of an existing transseries program. They do not settle general questions about transseries fields or every  $q$ -product regime. They do turn an explicitly unresolved error-analysis boundary in the inspected chapter into a collection of proved formulas and testable, reusable inversion tools.

## A Explicit bounds for finite computation

These estimates make the positive formulas numerically usable without pretending that a finite sum is an infinite one. Let  $p = 2M$ , let the positive weight satisfy  $0 < w_m \leq be^{-\lambda m}$ , and choose an integer  $N \geq p/\lambda$ . For a multinomial,  $\lambda = \alpha hx$ ; in the central case  $b = 2$  and  $\lambda = hx$ .

Since  $u^p e^{-\lambda u}$  decreases for  $u \geq N$ ,

$$\sum_{m>N} m^p e^{-\lambda m} \leq \int_N^\infty u^p e^{-\lambda u} du = \frac{\Gamma(p+1, \lambda N)}{\lambda^{p+1}}. \quad (\text{A.1})$$

Here  $\Gamma(a, z) = \int_z^\infty u^{a-1} e^{-u} du$  is the upper incomplete gamma function, a finite exponential-polynomial expression when  $a = p+1$  is a positive integer. The corresponding absolute remainder-tail bound is

$$E_m \leq \frac{h\zeta(p+2)}{2\pi^2} \left(\frac{h}{2\pi}\right)^p \frac{b\Gamma(p+1, \lambda N)}{\lambda^{p+1}}. \quad (\text{A.2})$$

For the omitted  $k > K$  terms at retained  $m \leq N$ , write  $z_m = (hm/(2\pi))^2$  and  $v_m = w_m z_m^M$ . Then

$$E_k \leq \frac{h}{2\pi^2} \frac{\sum_{m \leq N} v_m}{(p+1)K^{p+1}}. \quad (\text{A.3})$$

This is the integral test for  $\sum_{k>K} k^{-p-2}$ . The positive rectangular partial sum of (3.4), without its parity sign, is a lower bound; adding  $E_m + E_k$  is an upper bound. No cancellation of huge numbers is needed for the small remainder itself.

For the independent thermal reference (10.1), the bracketed kernel lies between zero and  $1/2$ , since  $e^z - 1 > z$  and its cotangent representation is positive after the first two terms are removed. Therefore the  $m > N$  tail is at most

$$\frac{be^{-\lambda(N+1)}}{2(N+1)(1 - e^{-\lambda})}. \quad (\text{A.4})$$

Finally, if the modular divisor sum is stopped after  $J$  terms,

$$0 < \mathcal{E}(h) - \sum_{n=1}^J \sigma_{-1}(n) r_h^n \leq \frac{r_h^{J+1} \{(J+1) - Jr_h\}}{(1 - r_h)^2}. \quad (\text{A.5})$$

All these are analytic inequalities. Outward rounding is still necessary when implementing their right-hand sides as rigorous numerical enclosures.

## B Coefficient construction and checks of normalization

The moments needed by the Padé construction can be evaluated without symbolic differentiation of large rational expressions. For  $n \geq 0$ ,

$$\text{Li}_{-n}(t) = \sum_{j=0}^n j! \left\{ \begin{matrix} n+1 \\ j+1 \end{matrix} \right\} \left( \frac{t}{1-t} \right)^{j+1}, \quad (\text{B.1})$$

where braces denote Stirling numbers of the second kind. To prove the identity, start with  $\text{Li}_0(t) = t/(1-t)$ , apply  $t d/dt$ , and use the Stirling recurrence. At  $t = 1/2$ , (5.5) becomes the explicitly rational finite sum

$$\mu_j = \frac{|B_{2j+2}|}{(2j+2)!} \sum_{k=0}^{2j} k! \left\{ \begin{matrix} 2j+1 \\ k+1 \end{matrix} \right\} \left( 2 - 3^{-(k+1)} \right). \quad (\text{B.2})$$

This is the formula executed in the script.

For a formal series  $f(s) = \sum_{j \geq 0} c_j s^j$ , a normalized Padé denominator  $q(s) = 1 + \sum_{j=1}^N q_j s^j$  for numerator degree  $L$  is found from

$$\sum_{j=1}^N q_j c_{k-j} = -c_k, \quad k = L+1, \dots, L+N, \quad (\text{B.3})$$

with  $c_j = 0$  for  $j < 0$ . The numerator coefficients are  $p_k = \sum_{j=0}^{\min(k,N)} q_j c_{k-j}$ . In our application  $c_j = (-1)^j \mu_j$ . Positive definiteness of the moment form gives the nondegeneracy needed in [theorem 5.2](#); the program additionally checks the relevant finite determinants exactly.

Several normalization traps are worth recording. The Bernoulli sum in  $B_M$  contains  $M$  terms, whereas the first omitted moment has power  $2M$ . The remainder sign is  $(-1)^{M+1}$ , not  $(-1)^M$ , because the tail enters as  $A_h(t^2) - 2A_h(t)$ . The next-term leading prefactor is *twice* the actual optimal remainder prefactor. The modular contribution to  $L_h$  is positive, because it is  $-\log(r_h; r_h)_\infty$ . Finally, the inverse approximation omitting that contribution is shifted by  $+\mathcal{E}/S'$  to leading order; interchanging the exact and approximate inverse reverses the sign.

## C Package contents and build instructions

The archive is self-contained and does not require the repository's shared notation file. It contains the editable `q_transition.tex`, its compiled PDF, this manuscript's verification script and recorded JSON output, a minimal dependency file, a provenance record, and a README.

Compile the document with a current TeX Live installation containing `newpx`, `tcolorbox`, `hyperref`, and `cleveref`:

```
pdflatex -interaction=nonstopmode -halt-on-error q_transition.tex
pdflatex -interaction=nonstopmode -halt-on-error q_transition.tex
pdflatex -interaction=nonstopmode -halt-on-error q_transition.tex
```

The bibliography is embedded, so BibTeX is unnecessary. To reproduce the checks, install the versions in `verification/requirements.txt` and run

```
python verification/verify.py
```

The program overwrites only its local `verification/results.json`. It makes no network request and does not alter the user's repository.



## References

- [1] V. Reshetnikov, *ProveIt*, repository snapshot 20301601562cf84173f2b503b0dc9c3fbf21cd0c. Companion manuscript, *Combinatorial Transseries and Their Inverses*, chapter “The singular transition  $q \rightarrow 1$ ”, labels `t3:prop:double-A`, `t3:eq:modularP`, `t3:eq:Sphase`, `t3:eq:crossover-all`. Inspected 29 September 2026. <https://github.com/VladimirReshetnikov/ProveIt/tree/20301601562cf84173f2b503b0dc9c3fbf21cd0c>.
- [2] V. Reshetnikov, *ProveIt*, “Series and transseries” and “Transseries: the polynomial-logarithmic calculus and its inversions”, repository README files at the same snapshot. The source inventory explicitly separates formal identities, partial analytic crosswalks, and absent assembled formalizations. See the source path in [section 11](#).
- [3] NIST Digital Library of Mathematical Functions, §4.36, “Infinite Products and Partial Fractions”, especially equations 4.36.3 and 4.36.4. Accessed 29 September 2026. <https://dlmf.nist.gov/4.36>.
- [4] NIST Digital Library of Mathematical Functions, §23.18, “Modular Transformations”, especially equation 23.18.5. Accessed 29 September 2026. <https://dlmf.nist.gov/23.18>.
- [5] C. Zhang, “A modular-type formula for  $(x; q)_\infty$ ”, *The Ramanujan Journal* **46** (2018), 269–305. <https://doi.org/10.1007/s11139-017-9967-5>.
- [6] V. Fantini and C. Rella, “Modular resurgence,  $q$ -Pochhammer symbols, and quantum operators from mirror curves”, *Letters in Mathematical Physics* **116** (2026), article 39. <https://doi.org/10.1007/s11005-026-02063-x>. Preprint <https://arxiv.org/abs/2506.08265>, version 2, 1 April 2026.
- [7] B. Simon, “The classical moment problem as a self-adjoint finite difference operator”, *Advances in Mathematics* **137** (1998), 82–203. <https://arxiv.org/abs/math-ph/9906008>.